

Project Cheap Adapter: one world, different coordinates

A bounded test of the Platonic Representation Hypothesis, converted into deployable linear adapters and a shared multi-encoder hub.

Project Documents

34 verdicts | 21 confirmed | 7 falsified | 4 inconclusive | 2 not meaningful

EXECUTIVE FINDING

The project confirms PRH in a narrow form: trained independent models share mainly local structure that linear maps make useful. It also bounds the claim: convergence is not global or universal, and visible geometry need not transfer.

RESEARCH PROGRAM

8 encoders | 2 shipped systems

1 Experiment A

GPT-2 vs Pythia-160M

CKA + layer-wise ridge on 10,000 passages tests whether independent language models share linearly accessible structure.

R² mean 0.737; peak 0.797

2 Experiment B

MobileCLIP to SigLIP 2

One 512x768 ridge matrix aligns phone and server image encoders and is exported as a 0.79 MB fp16 Core ML artifact.

R@5 = 95.3% of native ceiling

3 Cheap Adapter hub

Seven build it; one tests it

Seven encoders build the shared 512-d Hub. Held-out ConvNeXt is encoder eight and enters through its own ridge map.

N maps, not N(N-1)

EVIDENCE LADDER - STRONGER CLAIMS REQUIRE STRONGER TESTS

1	Shared neighbours	mutual k-NN	strong locally
2	Shared order	distance-rank rho	0.29 cross-modal
3	Shared structure	CKA / RSA	partial
4	Shared subspace	ridge + retrieval	88-97% of ceiling
5	Same geometry	rotation only	rejected: 0.038

The pattern is the result: models agree most on local neighbourhoods, less on global shape, yet become highly interoperable after a fitted linear stretch.

HEADLINE EVIDENCE

0.9183

MLP matches ridge in tested setup

0.592

vision adapter held-out R²

0.768

bge-SBERT raw shape rho

96.7%

ConvNeXt hub transfer / native

RELATION TO HUH ET AL.

Huh et al. linked alignment to scale and capability. Cheap Adapter tests the narrower mechanism: learned local structure is linearly usable across modalities and lineages, with held-out ConvNeXt at 96.7% of native. Limits: no universal or global identity; scaling is not monotonic beyond the tested range; the source-rank cliff is numerical.

From one adapter to the Cheap Adapter hub

The same scientific idea becomes an architecture: freeze every encoder, fit one linear entry map per space, and reuse downstream components across models.

SYSTEM DESIGN

PAIRWISE

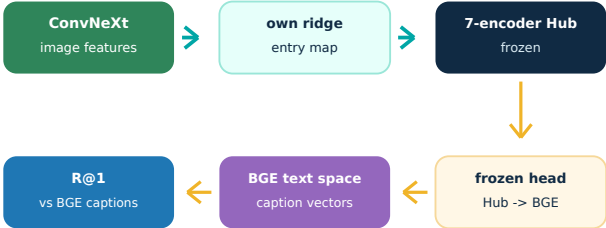
7 encoders -> 42 ordered maps

Adding one model requires maps to and from every existing model.

CHEAP ADAPTER HUB

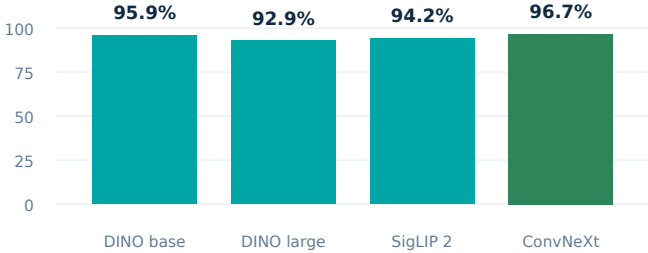
7 encoders -> 7 entry maps

7 encoders build the Hub; held-out ConvNeXt is encoder 8



MEASURED TRANSFER - HELD-OUT 1,000-ITEM BGE-M3 RETRIEVAL

% of a native bge-m3 retrieval head



WHAT TRANSFERRED

- Seven encoders build the Hub; ConvNeXt is the held-out eighth encoder.
- ConvNeXt fits its own closed-form ridge entry map; the Hub and caption head remain frozen.
- The flow ends in BGE text space and R@1 against BGE captions: 96.7% of native (adaptation, not zero-shot).

THREE COMPLEMENTARY RESULTS

MAP-FREE

rho = 0.290

Mean cross-modal distance-rank agreement before any map; bge-SBERT reaches 0.768.

COMPOSED

103.6%

Mean hub performance versus bespoke pairwise maps across all 42 ordered pairs.

DEPLOYED

0.79 MB

The fp16 MobileCLIP-to-SigLIP matrix reaches 95.3% of native R@5.

WHY THE INFERENCE IS CREDIBLE

Frozen models only closed-form maps are fitted	Held-out scoring 8,533 train / 1,000 test in the hub	Negative controls shuffle and random maps sit at chance	Nonlinear control MLP ties linear at 0.9183
--	--	---	---

PROJECT IN ONE PARAGRAPH

Two models do not share coordinates: direct cross-space retrieval is at chance without an adapter. Yet they encode related structure, and one phone-sized matrix makes it usable. What they share most strongly is local - nearby items agree more than the global shape. Coordinate frames are cheap to fix; semantic arrangement is what training buys. This does not imply one identical latent space.

Fractal probe: why shared reality matters

FractalDB tests a key PRH boundary: what structured training contributes without natural photographs, compared with matched random-init and ImageNet twins.

THREE TESTS, ONE CONSISTENT READING

G8

Visible geometry

Four map-free metrics reproduce nonzero cross-modal structure. Raw shape rho is 0.288 (0.329 without collapsed GPT-2); whitening lowers it to 0.062.

G9

Training creates geometry

Fractal training produces small but real shared geometry above matched random twins. Whitening reveals masked CNN structure, and the CNN signal survives PC1 removal.

G10

Geometry is not enough

That visible geometry does not become usable semantic retrieval through the frozen Hub: fractal models remain at the random floor, while ImageNet twins transfer strongly.

G9 - VISIBLE STRUCTURE RELATIVE TO MATCHED CONTROLS

Reality fraction R

0 = random architecture; 1 = ImageNet twin

Fractal CNN

0.09

Fractal ViT

0.21

Matched-norm CNN control: 0.067 fractal vs 0.036 random. The training effect survives preprocessing correction.

Robustness and uncertainty

- CNN image-vs-text R gap: 95% CI [-0.106, +0.100].
- ViT image-vs-text R gap: 95% CI [-0.053, +0.135].
- Corrected drop-1: CNN ordering survives; ViT falls below random.
- CNN-vs-ViT contrast is not interpreted because capacity differs.

G10 - FROZEN HUB/HEAD RETRIEVAL (R@1; OPERATIONAL FLOOR = 0.050)

ENCODER	TRANSFER	NATIVE	RANDOM TWIN	INTERPRETATION
Fractal CNN	0.014	0.014	0.012	at floor; no training-specific gain
Fractal ViT	0.020	0.022	0.017	at floor; no training-specific gain
ImageNet CNN	0.258	0.274	-	clear functional transfer
ImageNet ViT	0.204	0.208	-	clear functional transfer

PROJECT VERDICT LEDGER - 34 PRE-REGISTERED PREDICTION UNITS

21

CONFIRMED

Includes learned shared geometry, cross-modal and cross-lineage transfer, and fractal geometry above random.

7

FALSIFIED

Includes the stronger claim that visible fractal geometry must yield semantic transfer; it did not.

4

INCONCLUSIVE

Whether natural-world dependence differs systematically between image and text encoders.

2

NOT MEANINGFUL

Fractal percentage-of-native ratios, because both transfer scores lie below the 0.050 floor.

WHAT THE FRACTAL CONTROL ADDS TO PRH

Fractal training creates shared geometry above random, so convergence is learned rather than purely architectural. But retrieval remains at the random floor while ImageNet twins transfer strongly: shared natural-world experience is load-bearing, and visible geometry alone is not enough for recoverable semantic alignment.